

# InsightStream: Cloud Data Warehouse & Analytics

Building a Modern ELT Pipeline with Airflow, Snowflake Internal Storage, dbt, and Power BI

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# Project Overview

## Project Proposal

Overview of the project, objectives, and scope for building a robust, modern ELT pipeline for data analytics using Snowflake's native capabilities and industry-standard transformation tools, focusing on large-scale **Dataset Ingestion** and complex data modeling.

## Business Objectives

- Establish a comprehensive ELT pipeline for ingesting diverse **Datasets** directly into a cloud data warehouse.
- Enable advanced analytics and business intelligence through **Snowflake** and **Power BI** for data-driven decision making.
- Automate data workflows using **Airflow** and **dbt** to ensure data freshness, reliability, and scalability.

# Technical Objectives

- ✔ **Data Ingestion:** Ingest large-scale **Datasets (CSV, Parquet)** directly into **Snowflake Internal Storage** (Internal Stages) for raw data storage.
- ✔ **Data Transformation & Warehousing:** Utilize **dbt (data build tool)** to perform complex data cleaning, modeling, and transformation within Snowflake, ensuring modularity and automated documentation.
- ✔ **Workflow Orchestration:** Implement **Airflow DAGs** to manage and automate the end-to-end ELT process.
- ✔ **Business Intelligence:** Connect Snowflake to **Power BI** for creating powerful, interactive dashboards and enterprise-grade visualizations.
- ✔ **Monitoring & Automation:** Set up monitoring for Airflow DAGs and dbt runs, ensuring data quality and pipeline health.
- ✔ **Documentation:** Provide comprehensive documentation for the architecture, dbt models, and Power BI reports.

# Project Documentation Guidelines

| Item                                     | Student Deadline | Graduate Deadline |
|------------------------------------------|------------------|-------------------|
| Project Planning & Management            | 2/20/2026        | 2/3/2026          |
| Literature Review                        | 4/20/2026        | 3/1/2026          |
| Requirements Gathering                   | 4/20/2026        | 3/1/2026          |
| System Analysis & Design                 | 5/1/2026         | 4/3/2026          |
| Implementation (Source Code & Execution) | 7/10/2026        | 5/17/2026         |
| Final Presentation & Testing & Reports   | 7/17/2026        | 5/24/2026         |

# LITERATURE REVIEW

## Researching Best Practices for Modern ELT Pipelines

-  Techniques for ingesting large-scale **Datasets (e.g., CSV, Parquet)** directly into **Snowflake Internal Stages**, optimizing for performance and cost.
-  Best practices for data modeling using **dbt**, focusing on Modular SQL, Macros, and automated testing frameworks.
-  Effective use of **Apache Airflow** for orchestrating dbt jobs and data ingestion tasks.
-  Designing compelling and interactive dashboards using **Power BI** for high-impact business intelligence.
-  Strategies for ensuring **Data Quality and Governance** across the pipeline using dbt tests and Snowflake features.

# Requirements Gathering Details

## Functional

- Integration of diverse **Dataset sources**.
- Extraction of specific data points for business logic.
- Complex transformations using **dbt models**.
- Interactive analytical outputs in **Power BI**.

## Non-Functional

- Low **data latency** and high throughput.
- Robust **data security** within Snowflake.
- Cost optimization for cloud resources.
- System scalability and disaster recovery.

## Data

- Identification of all **Dataset formats**.
- Support for multiple data formats (JSON, CSV, Parquet).
- Handling varying **data volumes**.
- Defined frequency of updates for all sources.

# System Analysis & Design



Datasets



Snowflake Internal  
Storage



dbt Transformation



Power BI Analytics

Orchestrated by Apache Airflow

## System Architecture

A modern ELT approach where data is ingested directly into **Snowflake Internal Stages** from various **Datasets**. This eliminates external dependencies and reduces latency for raw data availability.

## Data Model (dbt)

Utilizing **dbt** to manage the transformation layers: **Staging** (cleaning), **Intermediate** (joining), and **Marts** (business logic), all optimized for analytical performance.

# Implementation: Tools & Technologies

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## Python

Core language for **Dataset processing** and Airflow DAG development.



## Snowflake

Cloud Data Warehouse utilizing **Internal Storage** for efficient raw data staging.



## dbt

Data Build Tool for SQL-based transformations, version control, and automated documentation.



## Apache Airflow

Workflow orchestration for scheduling and monitoring the end-to-end ELT pipeline.



## Power BI

Enterprise BI tool for creating interactive dashboards and data visualizations.



## Local/Cloud Datasets

Diverse data sources (e.g., CSV, Parquet) integrated into the pipeline for ingestion.



# Implementation Plan

## 01 Cloud Environment Setup

Configure Snowflake accounts, internal stages, and the Apache Airflow environment.

## 03 dbt Project Setup

Initialize dbt, configure profiles, and define sources.

## 05 Airflow DAG Development

Design DAGs to orchestrate ingestion scripts, dbt runs and tests.

## 02 Data Ingestion to Snowflake

Develop Python scripts to load **Datasets** and upload directly to Snowflake Internal Stages.

## 04 Transformation Layer

Write dbt models to transform raw data into analytical tables.

## 06 Power BI Dashboard Creation

Develop interactive dashboards connected to Snowflake.

# Final Presentation, Testing & Reports

## Testing Procedures

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### Unit Testing

Test individual Python scripts and **dbt models** for logic accuracy.

### Data Quality Testing

Use **dbt tests** (unique, not\_null, relationships) to ensure data integrity.

### Integration Testing

Verify end-to-end data flow from source **Datasets** through Snowflake/dbt into Power BI.

### Performance Testing

Evaluate **dbt run times** and Power BI query performance for scalability.

## Presentation & Reports

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### Final Presentation

Showcase project architecture, **dbt lineage**, and key analytical insights.

### Technical Report

Comprehensive documentation including **auto-generated dbt docs** and Power BI designs.

### Live Demonstration

A live walkthrough showing data flowing from source to interactive dashboards.

# Presentation & Reports Details



## Final Presentation

- ▶ Showcase the project's **modern ELT architecture**.
- ▶ Visualize **dbt lineage graphs** for data transparency.
- ▶ Present key analytical insights derived from **Power BI**.
- ▶ Discuss challenges overcome during implementation.



## Technical Report

- ▶ Comprehensive documentation of **design choices**.
- ▶ Inclusion of **auto-generated dbt documentation**.
- ▶ Detailed **Power BI dashboard designs** and logic.
- ▶ Cloud service configurations and security protocols.



**Live Demonstration: End-to-End Data Flow**

This project aims to deliver a robust, scalable, and modern ELT solution for data analytics, leveraging the power of **Snowflake**, **dbt**, **Airflow**, and **Power BI** to provide data-driven insights and support informed decision-making, with a focus on **large-scale Dataset processing**.

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# THANK YOU!

InsightStream Project Team